THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which observation indicates that a chemical reaction has taken place? 1

(a) change in temperature **(b)** change in colour
(c) evolution of a gas **(d)** all of these

2. The chemical formula of rust is: 1

(a) FeO **(b)** $Fe_2O_3 \cdot x\,H_2O$ **(c)** Fe_3O_4 **(d)** $Fe(OH)_2$

3. A balanced chemical equation is based on the law of: 1

(a) conservation of mass **(b)** conservation of energy
(c) constant proportions **(d)** multiple proportions

4. $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ produces a precipitate of colour: 1

(a) white **(b)** yellow **(c)** blue **(d)** black

5. A student states: “Burning of candle wax is only a physical change.” The correct response is: 1

(a) Correct
(b) It is only chemical
(c) Incorrect — melting is physical but burning is chemical
(d) Cannot be determined

6. Acids taste: 1

(a) sour **(b)** salty **(c)** sweet **(d)** bitter

7. Turmeric turns reddish-brown in a: 1

(a) acid **(b)** neutral solution
(c) salt solution of NaCl **(d)** base

8. Lactic acid is present in: 1

(a) orange **(b)** curd **(c)** tamarind **(d)** vinegar

9. A student states: “Methanoic acid in nettle stings is neutralised by an acid.” The correct response is: 1

(a) Correct **(b)** Incorrect — it is neutralised by a base
(c) It cannot be neutralised **(d)** Cannot be determined

10. Non-metallic oxides are generally: 1

(a) amphoteric **(b)** neutral **(c)** acidic **(d)** basic

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What happens when quicklime is added to water? Write the equation. 2

12. Balance: 2

(i) $H_2 + O_2 \longrightarrow H_2O$

(ii) $Mg + HCl \longrightarrow MgCl_2 + H_2$

13. Name two natural acid–base indicators. 2

14. What are alkalis? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Classify with reason: 3

(i) $2\,KI + Cl_2 \longrightarrow 2\,KCl + I_2$

(ii) $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$

(iii) $C + O_2 \longrightarrow CO_2$

16. How does a non-metallic oxide react with a base? Write the equation for carbon dioxide and calcium hydroxide, and conclude the nature of the oxide. 3

17. Explain with equations what happens when 3

(i) zinc is added to copper sulphate solution

(ii) sodium sulphate is added to barium chloride solution

(iii) magnesium is burnt in air.

OR

What is a redox reaction? Identify the oxidising and reducing agents in 3

(i) $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$

(ii) $2\,H_2S + SO_2 \longrightarrow 3\,S + 2\,H_2O$.

18. Describe an activity to show that all acids contain hydrogen ions but not all hydrogen-containing compounds are acids. 3

OR

Give the chemical name, formula and one use of 3

(i) washing soda

(ii) baking soda

(iii) bleaching powder.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** An iron gate rusts during the rainy season. 5

(i) What is rust and what conditions are needed for rusting?

(ii) Why does painting prevent rusting?

(iii) Give two other methods of prevention.

(iv) Why is corrosion a serious problem?

OR

(i) What are combination reactions? 5

(ii) Write equations for: burning of coal, formation of water from its elements, formation of slaked lime.

(iii) Why are these reactions exothermic?

20. **Case study:** In a school laboratory, students test solutions with a universal indicator: A turns red, B green, C blue and D violet. 5

(i) Arrange them in increasing order of pH.

(ii) Which is neutral?

(iii) Which is a strong base?

(iv) Which would react with magnesium to give hydrogen?

(v) Which could be distilled water?

OR

(i) What are salts? How are they classified as acidic, basic and neutral? 5

(ii) Classify: NaCl, Na_2CO_3, NH_4Cl with reasons.

END OF QUESTION PAPER — Total: 40 Marks